Fiscal Year:	FY 2005	**Task Last Updated:**	FY 06/22/2005
PI Name:	Sinha, Shantanu Ph.D.		
Project Title:	In vivo Stress-Strain Dynamics in Human Muscle		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Muscle Alterations and Atrophy Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	90095-1721	**Congressional District:**	30
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NSBRI
Start Date:	08/01/2001	**End Date:**	10/31/2005
No. of Post Docs:	2	**No. of PhD Degrees:**	2
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Edgerton, Reggie (University of California, Los Angeles) Elashoff, Robert (University of California, Los Angeles) Hodgson, John (University of California, Los Angeles) Lai, Alex (University of California, Los Angeles) Roy, Roland (University of California, Los Angeles)		
Grant/Contract No.:	NCC 9-58-MA00205		
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Task Description:	Muscle atrophy refers to the wasting or loss of muscle tissue resulting from lack of activity due to several reasons, including prolonged bed rest, limb suspension and immobilization due to disease and trauma, aging and spaceflight. Regarding space mission in microgravity environment, muscle atrophy and the accompanying neuromuscular disorders have been recognized as serious obstacles to be overcome for long-duration space exploration. Although many research projects have well documented morphological, biological and physiological changes in atrophied muscles using spaceflight and ground-based experimental models, it is virtually unknown what functional implications the changes in mechanical properties and structure of intramuscular tendinous tissues have. In addition, in order to investigate this issue in human muscles in vivo under the environment closer to physiological condition, it is required to develop a non-invasive measurement technique. For the last three years, our laboratory has developed, verified, and perfected a completely non-invasive measurement technique using velocity-encoded, phase contrast MR imaging combined with a custom-build MR-compatible muscle strength testing apparatus. This method enabled us to accurately quantify muscle volume and strength and strain dynamics of muscular-tendinous structures in human lower leg. Using this method, we examined the effect of a period of decreased mechanical load/suspension and the changes which occur upon recovery of the muscle to normal loading condition. For 13 subjects, atrophy was induced using a 4-week unilateral limb suspension (ULLS) model followed by 6-week rehabilitation. The ULLS model used was very effective for inducing localized atrophy, completely reversible and well-tolerated by the normal volunteers. Preliminary results of the previous ULLS study (performed in the prior years of the grant) showed that the decrease in muscle volumes following ULLS was proportionate to similar changes in relative strain distribution of the tendinous structures, suggesting that the structure and function are tightly coupled. Moreover, preliminary results of the most recent ULLS study (performed in 2004) demonstrated that absolute strain distribution (assessed with 10 and 20 % Pre-ULLS MVC) of tendinous structures changed following ULLS, possibly due to different motor unit recruitment strategy and altered mechanical properties of the structures. In addition, a different rehabilitation paradigm with moderate resistance training protocol (using loading condition less than subject's body weight) was tried out and found to be very effective. For the coming years, we would like to pursue better understanding of functional implications of architecture and mechanical linkages in atrophied muscles using MR Tractography on the top of current method. We strongly believe that future research study will provide valuable information to understand the structure and functional properties of normal and atrophied muscle and to develop more effective exercise regimen for astronauts.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The method developed herein, namely, using a non-invasive method of monitoring structural and functional differences between normal and atrophied muscles has a tremendous potential for Earth-based clinical use. Atrophy can result not only from micro-gravity as in the case of astronauts in space flights, but also from prolonged bed-rest or immobilization from casting. In these cases, the extent to which the muscle normalizes subsequent to rehabilitation is a very important clinical question, and can be addressed quantitatively by the method developed herein.
Task Progress:	In the reporting year, we carried out an experiment for four subjects investigating strain dynamics of muscular-tendinous structures of human lower leg in which muscle atrophy was induced using 4-week unilateral limb suspension (ULLS) and recovered normalcy throughout 6-week rehabilitation. We used exactly the same measurement technique (MR imaging and dynamometry) and experimental protocols (4-week ULLS and 6-week rehabilitation) as used in previous experiments except loading condition for the measurement of strain dynamics. The ULLS effectively induced muscle atrophy in terms of muscle volume and maximum isometric ankle plantarflexion torque. Strain dynamics of muscular-tendinous structures are under analysis. Preliminary results of the study will be presented in a conference. We will complete data analysis in near future and plan to publish in an appropriate scientific journal.
Bibliography Type:	Description: (Last Updated: 02/02/2007)
Articles in Peer-reviewed Journals	Finni T, Hodgson JA, Lai AM, Edgerton VR, Sinha S. "Mapping of movement in the isometrically contracting human soleus muscle reveals details of its structural and functional complexity Journal of Applied Physiology" 2128-2133 , Jan-2005
Articles in Peer-reviewed Journals	Finni T, Hodgson JA, Lai AM, Edgerton VR, Sinha S. "Nonuniform strain of human soleus aponeurosis-tendon complex during submaximal voluntary contractions in vivo Journal of Applied Physiology" 829-837 , Jan-2005
Articles in Peer-reviewed Journals	Finni, T., Hodgson, J. A., Lai, A., Edgerton, V. R. and Sinha, S. "Muscle Synergism during Isometric Plantarflexion in Achilles Tendon Rupture Patients and in Normal Subjects revealed by Velocity-Encoded Cine Phase-Contrast MR Radiology" Jan-2005
Articles in Peer-reviewed Journals	Sinha, S., Hodgson, J. A., Finni, T., Lai, A. M., Grinstead, J., Edgerton, V. R. "Muscle Kinematics During Isometric Contraction: Development of Phase Contrast and Spin Tag Techniques to Study Normal and Atrophied Muscles Journal of Magnetic Resonance Imaging" Dec-2004
Presentation	Sinha, S., Finni, T., Hodgson, J. A., Lai, A. M., Edgerton, V. R. "MR Phase Contrast Study of Differences in StructureFunction Relationship during Isometric and Passive Movement of Lower Leg " May-2004
Presentation	Sinha, S., Lai, A. M., Finni, T., Hodgson, J. A., Edgerton, V. R. "Assessment of Functional Recovery after Surgical Repair of Achilles Tendon Rupture from Estimates of Contraction Velocity, Muscle Volume and MVC " Nov-2003
Presentation	Sinha, S., Lai, A. M., Hodgson, J. A., Finni, T., Grinstead J and Edgerton, V. R. "An MR phase-contrast in-vivo study of Strain Distributions during Passive and Dynamic Movements of the Muscles of the Lower Leg " Nov-2004
Presentation	Sinha, S., Lai, A. M., Hodgson, J. A., Finni, T., Grinstead J and Edgerton, V. R. "Functional MR Characterization of Atrophied Human Triceps Surae Muscle following Four weeks of Lower Limb Suspension " Nov-2003
Presentation	Hodgson, J. A., Roiz, R., Finni, T., Lee, H. D., Edgerton, V. R., and Sinha, S. "Amplification of muscle fiber length changes in human soleus muscle-tendon complex " Sep-2004
Presentation	Lee, H. D., Hodgson, J. A., Finni, T., Edgerton, V. R., and Sinha, S. "Strain distribution of in vivo human triceps surae during passive and active dynamic movements " Sep-2004

Presentation	Sinha, S., Finni, T., Hodgson, J. A., and Edgerton, V. R. "Structure-Function Correlation of the Human Soleus Muscle " Nov-2004